Acquirium: Toward Interoperable Data Driven Applications in Water Treatment Systems

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Abstract

Water treatment systems face increasing pressure for automation driven by decentralization trends and regulatory requirements, yet deploying the necessary data-driven processes remains difficult due to fragmented data sources, unstructured metadata, and lack of interoperability across facilities. We present **WaTr**, an ontology designed for water treatment operations that captures fine-grained treatment processes, substance flows, and external references to heterogeneous data sources, and **Acquirium**, a co-designed data management platform that enables metadata exploration without requiring expertise in semantic query languages. WaTr recasts and significantly extends existing building ontologies to support a wide array of water treatment technologies and concepts. Acquirium supports a novel dataframe-style API for progressive graph query refinement and a lightweight "soft-sensor" abstraction for deploying portable analytics. We demonstrate that Acquirium reduces application code by up to 4 times compared to existing platforms, while scaling to hundreds of concurrent processes and complex soft-sensor compositions with sub-second end-to-end latency. These contributions lower the barrier to developing interoperable, data-driven applications for water treatment facilities.

CCS Concepts

• **Information systems** → **Information retrieval**; **Database design and models**.

Keywords

Semantic Interoperability, Ontologies, Knowledge Graphs, Smart Infrastructure, RDF, Data Management

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1 Introduction

Water and wastewater treatment accounts for approximately 2% of total U.S. energy consumption and is carefully instrumented and regulated as critical infrastructure [72]. Further, energy costs represent up to 30% of a water utility's operations and maintenance costs [72]. While substantial literature exists on applying data to manage and optimize building subsystems like HVAC and electrical, water treatment infrastructure remains underserved by data-driven applications at the building, campus, and municipal scales. These applications can reduce costs, improve reliability of decentralized systems, and enable coordination with other sectors (e.g., building and energy systems).

The growth in onsite water treatment and reuse systems has been driven by regulation (e.g., mandating onsite wastewater reuse for new buildings > 100,000 sq. ft. [61]), cost savings opportunities associated with high connection fees and water/wastewater tariffs, and an awareness of the resiliency and sustainability benefits of reusing water onsite. Decentralization also yields faster deployment during droughts [45, 81], reduced capital expenditures, lower conveyance costs [45], and value to rural communities lacking density for centralized infrastructure [9, 30, 56]. By enabling fit-for-purpose water use, these systems support broadly shared societal goals related to securing water and food supply chains [26].

While small scale systems can complement large, centralized water infrastructure; their design and operation introduces unique data management, process control, and verification challenges. Centralized treatment facilities rely on constant operator oversight during operation [55]. These operators are often assisted by automation and set-point control of specific unit processes requiring expensive heuristic tuning. In contrast, the economic feasibility of small-scale systems depends on rapid commissioning and remote supervision, control, and alarming to identify when onsite maintenance is required [45]. Because small treatment plants cannot afford dedicated technical staff or expensive consultants [36], monitoring and automation are essential components of their operation and maintenance to achieve potential cost benefits [26]. The heterogeneity and variability of water sources and treatment processes further reinforce the need for automation. Rapid plant commissioning and adoption of automated operation is hampered by current digital infrastructure and data management practices [19].

Data collection accounts for approximately 40% of effort in water-sector research and analysis projects [1, 74]. The roots are structural: operational data is fragmented across heterogeneous sources, such as Supervisory Control and Data Acquisition (SCADA), historians, laboratory spreadsheets, and manual logs; metadata describing equipment layout and instrumentation resides in P&IDs and PFDs that are rarely machine-readable and site-specific naming conventions require expert interpretation. As a result, digital solutions are one-of-a-kind and work is duplicated at each new facility due to a lack of interoperability [28].

Semantic ontologies can eliminate data access problems [10, 11, 22, 29], but two barriers prevent their direct application to water treatment. First, while several semantic models address aspects of the water domain (§2.2), none captures the operational concepts required for treatment analytics, including equipment connectivity, treatment processes, substance flows, and references to heterogeneous data sources. Second, existing platforms assume users are fluent in specialized query languages and familiar with the graph structure.

This paper presents WaTr, an ontology designed to support water treatment system operations, and Acquirium, a co-designed data management platform that enables incremental metadata exploration. WaTr extends the 223P ontology [5], inheriting its framework for equipment connectivity while adding vocabulary for treatment processes, chemical media, and heterogeneous data sources including laboratory measurements. Acquirium (first framed in our previous poster [60]) provides a dataframe-style API that supports progressive query refinement, allowing users to explore plant metadata interactively rather than writing exact declarative queries. These support the development of data-driven applications in resource-limited water treatment facilities.

We make the following contributions:

- **Challenge taxonomy:** We classify the data and metadata management challenges that water treatment facilities face when implementing data-driven applications, mapping eleven representative applications to specific bottlenecks (Table 1). These challenges disproportionately affect smaller plants with constrained resources.
- **WaTr ontology:** We develop an ontology for water treatment system operations, extending an existing building ontology to fully or partially cover the majority of SDWIS treatment objectives and processes [71] and WaterTAP flowsheets [8, 77].
- **Acquirium platform**: We design a data management framework featuring incremental metadata discovery, unified access to heterogeneous data sources (SCADA, historians, CSV files, MQTT streams, laboratory measurements), and a lightweight application abstraction for deploying portable analytics and composable soft-sensors.

Both WaTr[1] and Acquirium[2] are **open-source**.

2 Background and Related Work

2.1 Water Treatment Systems

Operational data in water treatment systems (WTS) spans diverse temporal resolutions, sensing modalities, data formats, and acquisition processes, while related metadata is scattered across numerous sources. WTS share a structural foundation with building mechanical systems — networks of connected equipment instrumented with sensors — but additionally require modeling treatment trains, chemical processes, media, and heterogeneous data modalities like laboratory samples.

WTS treatment trains consist of multiple unit processes connected in sequence. For example, wastewater treatment commonly includes grit removal, primary clarification, aeration, secondary clarification, sludge digestion, and disinfection [65]. Piping and Instrumentation Diagrams (P&IDs) provide detailed representation of equipment, pipes, valves, and connections, and the placement of sensors and actuators with identifiers or network names.

SCADA systems collect sensor data (e.g., conductivity, flow rate, volume, pH) at varying frequencies. Other quantities, such as volatile suspended solids, often require laboratory analyses [41]. Additionally, observations such as unusual odors or maintenance actions not recorded by automated systems are captured by manual logbook entries. This fragmentation of data is a key barrier to deploying data-driven applications. Moreover, some measurements cannot be obtained using existing system components and must instead be estimated from available sensor data. These estimates, known as *soft-sensors*, are generated using data-driven or model-based tools and enable continuous monitoring without additional physical instrumentation.

2.2 Knowledge Graphs and Semantic Metadata

Knowledge graphs (KGs) are directed, labeled graphs that represent typed entities (nodes) and the relationships between them (edges). An **ontology** specifies the classes of entities, the relationship types that may connect them, and their intended meaning [14, 57]. The graph representation decouples *system metadata* (assets, connections) from the *data produced by the system* (time series, lab results, logs), allowing applications to bind to semantics rather than to site-specific naming or ad hoc identifiers [11, 59].

We adopt the Resource Description Framework (RDF) [46] formalization of KGs which represents nodes and edges with globally unique identifiers that can be shared, merged, and extended across organizations and software stacks. RDF benefits from a mature ecosystem of graph databases and libraries, and a standard declarative query language (SPARQL [67]) for retrieving metadata based on graph structure and semantics.

Ontologies like Brick [11], ASHRAE 223P [5], and RealEstateCore [27] standardize the representation of system structure, assets, and sensing infrastructure by defining hundreds of concepts to support smart building applications.

[1]https://github.com/DataDrivenCPS/water-ontology
[2]https://github.com/DataDrivenCPS/acquirium

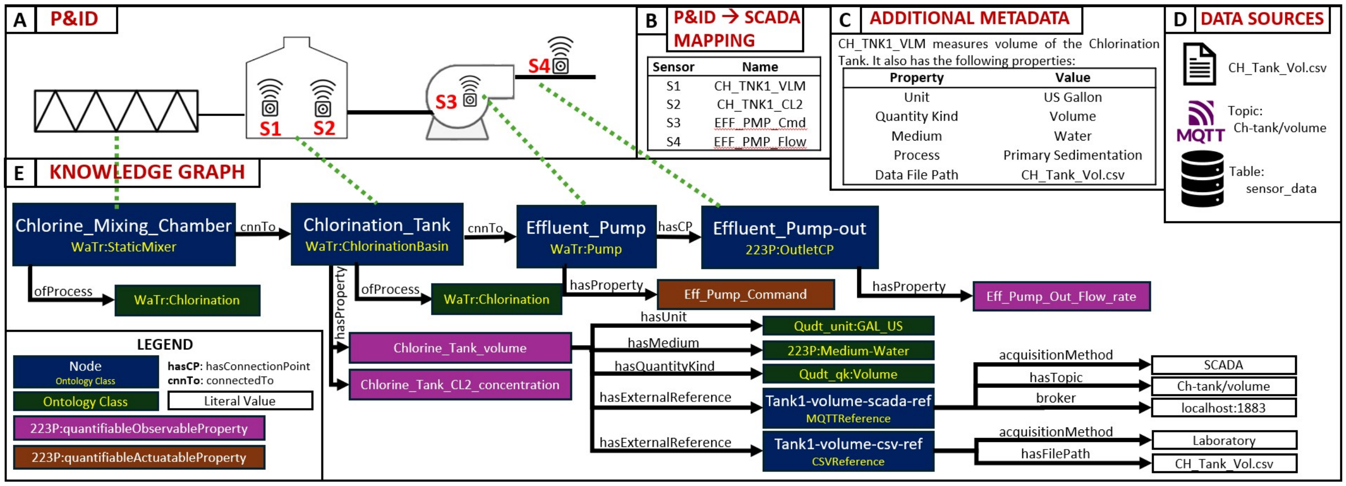


Figure 1: A P&ID snippet (A) shows a system composed of several unit processes and equipment. The subsystem has four sensors with SCADA tags (B). Sensors have additional metadata: unit, quantity kind, medium, process, data source, and external reference (C), none of which reside on the P&ID. Sensor data is stored across multiple files and systems (D). Our proposed knowledge graph (E) unifies all the metadata and links to external data sources.

2.3 Ontologies in Water

Other semantic models have been proposed in the water domain. [51] presents a KG model for capturing treatment process knowledge and cataloging predictive models, but the lack of underlying ontology limits cross-facility standardization. SAREF4WATR [25] is an ontology for representing water distribution systems from a utility perspective, with classes such as `Lake`, `WaterMeter`, and `TreatmentPlant`. [32] extends SAREF with water distribution network assets, smart metering, and consumer sociotechnical concepts. Neither [25] nor [32] focus on representing facility treatment trains. WAM-ONTO [80] is an asset management ontology for treatment plants that aims to preserve design knowledge across the facility lifecycle, however they do not focus on operational data access. The Internet of Water [66] initiative integrates water data across agencies using shared terminology and geospatial metadata for water features (lakes, rivers, etc.). [47] proposes a metadata framework for environmental model sharing. PyPES [18] proposes an object-oriented Python schema for water data management, modeling equipment, connections and data sources, but lacks a formal ontological layer to achieve standardization and interoperability.

To our knowledge, WaTr is the only ontology that focuses on WTS *operation*, including modeling diverse unit processes, detailed topology of treatment trains, and references to heterogeneous data streams. No other work provides the external reference mechanism for data sources that WaTr offers: rather than storing time-series values directly in the graph, WaTr allows applications to retrieve data from high-throughput, specialized time-series databases.

2.4 Data Management Platforms

Several data management platforms have been proposed to link semantic models to operational telemetry for portable data-driven applications. Mortar [22] couples a metadata model with time series storage and uses SPARQL to expose a unified interface that binds metadata to time series identifiers. Chrontext [10] enables querying a metadata model and telemetry in a single workflow. Energon [29] targets portability through a declarative query language, EnergonQL, for both ontology traversal and data retrieval. These systems require fluency in RDF, SPARQL, and the structure of the site's KG, since an exact query is needed to return results [13]. The scale and interconnectedness of entities make learning each new graph a substantial task, and zero-result queries offer no guidance for debugging. Here, we explore how to design a system that abstracts away from the underlying graph structure and query language, so that users can retrieve data by referencing domain entities and their relationships rather than writing complex graph queries.

2.5 Running Example

Figure 1 illustrates a chlorination subsystem instrumented with four sensors and control points (S1–S4). The P&ID captures the physical structure and instrumentation, but sensor metadata such as units, quantity kinds, associated processes, and data sources are maintained separately. The measurements themselves are similarly fragmented making contextual reconstruction dependent on site expertise. The KG integrates metadata from the P&ID, SCADA system, and data historian into a single standardized representation.

3 Data Challenges in Water Treatment

Many data driven applications have been proposed for WTS, spanning monitoring, optimization, and decision support. These approaches improve operational efficiency while reducing energy consumption, costs, emissions, and regulatory risk. This is increasingly important as more stringent requirements on greenhouse gas emissions [15] and water quality [50] are expected to increase the energy intensity of water and wastewater treatment. Deploying these applications in practice remains difficult due to the data management challenges faced by operators and engineers.

Table 1: Representative apps for WTS with data/metadata requirements and key challenges encountered in practice (§ 3).

#	Application	Required data and metadata	Encountered challenges
A1	Analysis of multi-source and fragmented datasets [58, 64]	Measurements from historian/DBs, lab spreadsheets, timestamps, units and quantity kinds, entity and location context	C-1.1; C-1.2; C-1.3
A2	Threshold based sensor monitoring [6, 82]	Sensor measurements (MQTT/SCADA), thresholds, units, entity and location context, alarm routing and notification information	C-1.2; C-1.4
A3	State transition and event detection (e.g., backwash cycle) [63]	Sensor measurements (historian/DB), entity and location context, event definitions and state transitions	C-1.2; C-1.3; C-2.1
A4	Automated fault detection and isolation [4, 24, 31, 33, 37, 42]	Sensor measurements (historian/DB), entity and location context, units, mediums, quantity kinds, alarms and logs	C-1.1; C-1.2; C-1.4; C-2.1
A5	Sensor drift detection and correction [2, 79]	Sensor measurements (historian/DB), calibration records, units, measurement type and range, lab spreadsheets, logs	C-1.1; C-1.3; C-1.4
A6	Data validation and quality assessment [2, 4]	Sensor measurements (historian/DB) and lab spreadsheets, expected ranges and units, entity and location context, sampling intervals	C-1.2; C-1.3; C-1.4
A7	Prediction of adsorbent performance for pollutant removal [3]	Sensor measurements (historian/DB) and lab spreadsheets, adsorbent properties	C-1.1; C-1.3; C-1.4
A8	Automatic detection of newly installed sensors [44]	Sensor measurements (historian/DB) and lab spreadsheets, entity and location context, device identifiers, network and SCADA tags	C-1.2; C-2.1; C-2.2
A9	Optimal sensor placement for flow monitoring [39, 75, 76]	Entity and location context, Sensor measurements (historian/DB), operational constraints and objectives	C-1.2; C-1.3; C-2.2
A10	Digital twins for water treatment facilities [12, 51, 70]	Entity and location context, simulation models, Sensor measurements (MQTT/SCADA), control signals and setpoints	C-1.3; C-1.4; C-2.2
A11	LLM based question answering over operational data [49]	Entity and location context, Sensor measurements (historian/DB), units, mediums and quantity kinds, logs and retrieval indices	C-1.1; C-1.2; C-1.3

Large plants mitigate these challenges through dedicated site experts and commercial software, increasing costs and limiting broader adoption [36]. Smaller plants, including building scale and rural facilities, may lack the required personnel and engineering support for expensive subscriptions and consultant-driven deployments on which existing solutions rely. Table 1 summarizes representative applications and maps each to specific data management challenges.

C-1 Data–Metadata Access and Discovery Challenges

C-1.1 Data Multimodality WTS applications require both operational measurements and contextual information about how the plant is built and operated. WTS data span multiple modalities including numerical time series, categorical metadata, and unstructured records such as text logs, images, and calibration documents [18, 74]. Access and discovery remain difficult because data is fragmented across many sources and nonstandard formats (e.g., manual logbook entries) [34].

C-1.2 Unstructured Metadata Sensor locations, equipment layout, process details, and operator notes (metadata) often live in different systems and formats than time series and laboratory measurements (data) [74]. Equipment layout and instrumentation are documented in P&IDs (e.g. Fig. 1-A) while process descriptions often reside in PFDs; these are not machine-readable. Data streams are commonly labeled with ad hoc naming conventions (e.g. Fig. 1-B) with interpretation requiring site-specific data dictionaries or expert knowledge.

C-1.3 Data Fragmentation Measurements are fragmented across systems. Sensor telemetry may be published through Message Queuing Telemetry Transport (MQTT) streams, historian data may reside in relational databases, SCADA systems may expose proprietary APIs, and lab measurements are often recorded manually in Excel or CSV files (e.g. Fig. 1-D). This fragmentation makes it difficult to locate all measurements relevant to a task and to reconcile duplicates, gaps, and inconsistent representations [58].

C-1.4 Data Quality Plants need reliable, high-quality data to inform operational decisions and satisfy regulatory requirements [78]. In practice, quality is degraded by sensor aging, calibration drift, communication failures, and missing data. Validation, outlier detection, and reliability checks are essential for safe and effective operation of water systems [4].

C-2 Plant Evolution and Interoperability Challenges

C-2.1 Interoperability Many monitoring and analytics requirements are shared across plants, but the supporting engineering artifacts are not standardized. While many P&IDs use ANSI/ISA 5.1 standardized symbols, the diagrams themselves are typically not machine readable [62]. Plants rarely produce these diagrams in modeling software with structured outputs. As a result, extracting actionable information (e.g., identifying which sensor measures the level of a specific tank) usually requires manual interpretation by a site expert, which hinders portability of software across facilities and raises software development costs [1, 74].

C-2.2 Plant Evolution Equipment, instrumentation, and operating configurations evolve, forcing updates of design documents

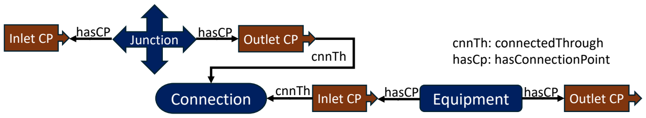


Figure 2: Primary concepts and relationships used to express connectivity and topology of equipment.

(P&ID, PFD, etc.) and manual modifications to data pipelines. This maintenance burden grows as plants evolve: each physical change must be propagated across documentation, simulations, and digital twin models [35].

4 Water Treatment Metadata Ontology

Applications for WTS require information about the topology of mechanical systems, their chemical processes, input and output media, and the identity and location of the data they generate. This *metadata* is fragmented across multiple resources (P&IDs, PFDs, data historians, SCADA systems). Each capture different facets of a system's design and operation, often with conflicting identifiers and labels. We propose a metadata ontology for WTS which unifies plant metadata into interoperable machine-readable graphs.

4.1 Ontology Overview

The WaTr ontology extends the ASHRAE 223P standard [5], which defines machine-readable semantic models for automation and control data in building systems, with modeling constructs for the water treatment domain. WaTr inherits: an existing framework for expressing equipment connectivity and topology; integration with the QUDT ontology [21] for units of measure and quantity kinds; compatibility with existing tools for graph validation (SHACL) and querying (SPARQL); and future interoperability with buildings that also adopt semantic ontologies. WaTr introduces and extends concepts where the base 223P vocabulary is insufficient for water treatment (Table 2).

WaTr uses four concepts to model substance flow information traditionally captured in PFDs and P&IDs (Figure 2): **Equipment** is a catch-all class for connected assets participating in the treatment train, and may have any number of **Connection Points** (CPs). Each CP has a direction (`Inlet`, `Outlet`, or `Bidirectional`) and associates with equipment, **Junction**s, or **Connection** objects (e.g., pipes, ducts). Connections and CPs are characterized by a **Medium**: anything that flows or permits the flow of energy or information. WaTr additionally describes mixtures composed of constituent **Substances** at known or unknown concentrations.

Following Brick's design principles [11], equipment classes are organized into a hierarchy reflecting functional role rather than topological interface; an `AerationBasin` and `ChlorinationBasin` may share identical CPs but perform distinct treatment. This organization mirrors how domain experts group devices in practice.

4.2 Processes and Measurements

WaTr differentiates between what a unit process accomplishes versus how it is constructed through the concept of a **Process**. The process enacted by a unit of equipment is defined by the `WaTr:hasProcess` property, which references process types such as sedimentation, UV disinfection, or chemical oxidation. This distinction allows consumers of a WaTr graph to query for all units performing a particular treatment function (e.g., "all equipment performing disinfection") regardless of their specific physical implementation (e.g., UV reactor vs chlorine contact chamber).

A **Property** represents an attribute of a feature of interest, such as outlet water temperature or chemical concentration in a tank. Properties link to topological elements (pipes, outlets, pumps) and can also attach to media or substances to express concentrations; this can model chemical dosing and pollutant monitoring.

Properties can be *observable* or *actuatable* depending if it can be sensed or controlled, and *quantifiable* or *enumerated* if its values are numerical or from a fixed set. Quantifiable properties are also described by a **Quantity Kind** and **Unit** defined by the QUDT ontology [21]. Quantity kinds are independent of any system of measurement, and include concepts like temperature, and concentration. Enumerated properties may be described using **Enumeration Kinds** which define enumerated values a property may take (e.g. on/off).

WaTr properties also carry data quality metadata including their expected range, numerical and temporal resolution, drop rates during collection, and any applied processing techniques. We store these attributes in the graph to enforce consistency with the direction, medium and/or substance of the many observable flows in the system.

Table 2: Counts and examples of concepts defined in WaTr.

Category	Count	Examples
Unit process classes	74	Sedimentation, UV Disinfection, Chlorination
Equipment classes	106	StaticMixer, AerationBasin, Clarifier, Membrane
Substances, media	32	Mixed-liquor suspended solids, Brine

4.3 External References

Properties link the topological view of a treatment train to external data through **External Reference** objects. Each reference carries the parameters needed to access another system. This could be a SCADA tag, publish-subscribe topic, API endpoint, database key, or some other identifier. Following `hasExternalReference` from a property lets applications locate data (Fig. 3; instance in Fig. 1-E).

4.4 Validation

WaTr leverages SHACL [68] to automatically catch modeling errors before applications consume them that would otherwise require expert review. Rules cover unit/quantity/property consistency, medium propagation through connections and equipment, and minimum required metadata.

5 Data and Metadata Management Platform

We develop **Acquirium**, a data management system designed to unify data access and processing across fragmented resources and annotate that data with rich, standardized graph metadata. Acquirium implements a novel dataframe-inspired API that supports

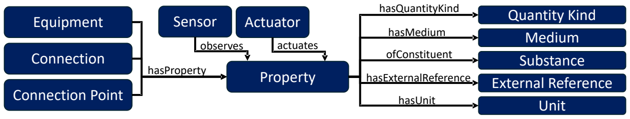


Figure 3: The primary concepts and relationships used to define a Property.

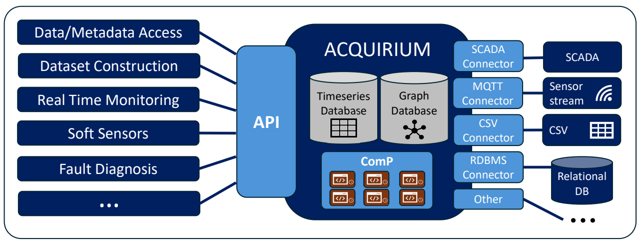


Figure 4: Acquirium system overview. Applications access data and metadata through a shared API which abstracts data access and retrieval.

data/metadata discovery without the need for complex graph query languages; this supports in-platform batch analytics and real-time monitoring workflows. These features enable interoperable applications that reduce manual reconfiguration effort while supporting efficient automation for resource-constrained water environments.

5.1 System Architecture

Acquirium (Figure 4) integrates a graph database storing plant metadata models (KGs) with a time series database storing telemetry. Acquirium uses an array of data connectors, similar to sMAP [20], that abstract data access behind a flexible pull/push interface to SQL databases, SCADA systems, file shares, and streaming sources (e.g., MQTT). A plant's KG embeds *external references* to these connectors (Fig. 1-E), linking each measurable point to one or more concrete data locations. Making source bindings explicit and queryable in the metadata layer lets each measurable point carry multiple external references, so data about the same entity from different sources can be joined through references to the same property in the KG.

5.2 Incremental Query Interface

Although graph-based metadata supports expressive WTS modeling, the query languages traditionally required to access it are complex [13, 73]. We define a programmatic dataframe-style interface providing *progressive* metadata and time series access, allowing operators and developers to build applications without learning RDF or SPARQL.

Traditional dataframes compose relational transforms over bags of named tuples, sitting between a SQL table and a spreadsheet [54]. We adopt their key property: *incremental refinement*, where users inspect results at each step and decide how to proceed. In contrast, SPARQL requires a complete graph pattern before execution; a subtly incorrect query can return an empty result set with no indication of the mismatch. Debugging queries requires understanding the graph well enough to identify the error, presenting a circular dependency that obstructs exploration.

Our abstraction treats queries as composable, inspectable objects: users begin with a simple selection (e.g., "find all tanks"), examine results, then extend to include related entities (e.g., "and their downstream pumps"). At each step, the current matches are visible and provides feedback. This shifts the burden from *knowing the graph structure in advance* to *discovering it through interaction.*

5.3 Core Abstractions

Let G represent the union of the plant metadata and its dependency graphs (i.e. ontologies) and let X denote the collection of all data records stored in the time series database of Acquirium (including time series samples, lab records, and logs). We define two core objects that mediate access to G and X: the **Query** object for metadata discovery and the **Data** object for value retrieval.

Query Objects. A Query object Q represents a graph pattern to be matched against G. The pattern consists of *node constants* (representing nodes in G), *node variables* (representing entities to be discovered) and *edge constraints* (representing relationships between those entities). Query objects are constructed incrementally through method chaining:

- `find_entity(_class=...)` introduces a new node variable constrained to instances of the specified class.
- `find_related(_class=..., _dir=...)` extends the pattern by adding a new node connected to an existing node via a path in the specified direction (upstream, downstream, or any).
- `filter(...)` adds constraints to an existing node variable without introducing new variables.

For usability, Acquirium permits natural-language-like class names and point descriptions and resolves them to ontology classes and candidate nodes through a text matching layer.

Evaluating Q against G yields a binding table M_Q where each row is one match of the pattern and each column is a node variable: $\texttt{eval}(Q, G) \rightarrow M_Q$. (e.g., Fig.5-`Out[1]`).

Crucially, evaluation is *lazy*: the Query object accumulates constraints without executing until the user explicitly requests results (e.g., via `head()`). We currently translate the object into SPARQL for evaluation, but plan to explore alternatives that better exploit incremental composition.

Figure 5 illustrates this workflow: Line 3 selects all `Tank` entities, Line 6 extends the pattern to include downstream pumps, and `head()` on Line 8 triggers evaluation, letting the user verify matches before proceeding.

Data Objects. A Data object D bridges the metadata (G) to the value (X). Given a Query Q whose matches include measurable properties, and a time context T (e.g., an interval $[t_{\text{from}}, t_{\text{to}}]$), a Data object retrieves values by: (i) identifying the external references associated with each matched property in M_Q, (ii) invoking the appropriate connector to fetch values from the referenced source over T, and (iii) joining the retrieved values with the metadata context from M_Q. We write this `materialization` abstractly as:

$$\texttt{mat}(D_{Q,T}) \;\rightarrow\; \{(m, v) \mid m \in M_Q,\; v = \texttt{read}(\texttt{ext}(m), T)\}$$

where $\texttt{ext}(m)$ returns the external reference(s) for the property in match m, and $\texttt{read}(r, T)$ fetches data from reference r over time

```
In [1]:
1 from acquirium import Acquirium
2 acq = Acquirium()
3 q1 = acq.find_entity(_class="tank")
4 q1.add_log(observation=("2024-02-01","2024-02-02"),
5            msg="Tanks are inspected, all is good!")
6 q2 = q1.find_related(_class="pump" , _dir="downstream")
7 # Tank --(*)--> Pump
8 q2.head()
```

```
Out [1]:
| Tank                    | Pump               |
|-------------------------|--------------------|
| Chlorination_Basin      | Effluent_Pump      |
| Diurnal_Flow_Box        | Flow_Return_Pump1  |
```

```
In [2]:
9  q3 = q2.find_related_data(_unit=["GAL_US","GPM"]
10 q3.dataframe(_from="2024-01-01",_to="2024-12-31").head()
```

```
Out [2]:
| Time                 | Volume  | Flow  |
|----------------------|---------|-------|
| 2024-01-12 00:00     | 150     | 12    |
| 2024-01-12 00:01     | 150     | 15    |
```

Figure 5: Example notebook showing incremental metadata querying and time series access.

context T. A Data object represents (i) SCADA and sensor telemetry as time series, (ii) laboratory measurements with richer time semantics (e.g., sample time versus record time), and (iii) log and annotation records with record time and an observation period, preserving each modality's time semantics behind a unified interface.

In Figure 5, Line 9 extends the query to data points filtered by unit, and Line 10 materializes a dataframe over the specified window. The resulting dataframe joins the semantic context (which tank, which pump, what property) with the numeric values, enabling analysis without manual tag lookups or source-specific code.

Handling Multiple Sources. Data objects support selection between multiple external references (e.g., both a SCADA tag and a laboratory measurement for chlorine concentration). Users can filter by acquisition method (e.g., `kind="lab"`) to retrieve values from a specific source, or retrieve all sources and reconcile them in application logic. This design directly addresses the data fragmentation challenge (C-1.3) without forcing a single canonical source.

5.4 Computed Properties

Many WTS applications such as soft-sensors require values derived from other measurements; e.g., computing energy intensity from flow rate and power consumption. These **computed properties (ComP)** are first-class entities in Acquirium. A ComP is defined by: (i) an *input query* Q_{in} specifying the source properties required for the computations, (ii) a *transform function* λ that maps input values to output values, and (iii) one or more *output property* nodes in the KG where results are stored. Each ComP caches Q_{in} result until the graph changes, avoiding repeated graph query execution across

```
1  from acquirium import Acquirium, App, Output
2  class Chlorine_Level_Warning(App):
3    name = "chlorine_warning"
4    app_type = "threshold"
5    outputs = [{ "kind": "trigger"}]
6    def build_query(self, acq: Acquirium):
7      return (acq.find_entity("Cl2 Basin")
8                .find_related_data(unit="Mg/L"))
9    def run(self, ctx) -> list[Output]:
10     df = ctx.query.data.latest(shape="wide")
11     msg = "HIGH" if df["Concentration"]>4 else "LOW"
12     return [Output.trigger(url="a.com/alert",msg=msg)]
13
14 acq = Acquirium()
15 acq.register_app(Chlorine_Level_Warning())
16 acq.run_app("chlorine_warning", interval=10)
```

Figure 6: Example ComP for monitoring Chlorine level.

runs. New values in source properties can trigger the transform to update the computed property with low latency (§ 6). Formally: $\lambda : \mathtt{mat}(M_{Q_{\text{in}}}, D_{Q_{\text{in}},T}) \longrightarrow X'$ where X' denotes the derived records written back to the timeseries store under the computed property's identifier.

Because ComPs are represented as `Property` nodes in the KG, they inherit the same metadata annotations as physical measurements: units, quantity kinds, media, process associations. The KG records provenance: each ComP links to its source properties via `dependsOn` relationships to trace how a value was derived. As a result, queries and applications can consume ComPs identically to measured ones, without needing to distinguish their origin.

In addition to timestamped values, ComP can trigger HTTP requests, MQTT publishes, and other external effects. ComPs can express one-to-one, many-to-one, or many-to-many mappings by defining more than one output property. ComPs are also *composable*, meaning they can serve as inputs to other ComPs. § 6 evaluates the scalability of composed ComPs.

Example: Figure 6 shows an example monitoring ComP that queries for chlorination basins, and retrieves data with unit mg/L. The ComP reads the latest values (line 10) and triggers a warning if the conditions exceed a threshold (lines 11-12), calling an external API. The App is then registered and executed on a schedule (lines 15 to 16), enabling periodic monitoring without manual data preprocessing.

5.5 Addressing Data Management Challenges

Unified access to heterogeneous sources (C-1.1, C-1.3). Acquirium abstracts data access behind connectors for SQL databases, SCADA, file and streaming sources (e.g., MQTT). Each measurable property can have multiple external references, so the same quantity (e.g., chlorine concentration) can be retrieved from both a SCADA tag and a laboratory spreadsheet. Users select sources by acquisition method (e.g., `kind="lab"`) or retrieve all sources for reconciliation.

Structured metadata without SPARQL (C-1.2). The incremental query interface and text matching layer allow users to discover equipment, connections, and measurement points without writing declarative graph queries or memorizing ontology terms. This lowers the barrier for operators and engineers who are unfamiliar with RDF or SPARQL.

Logs and annotations as first-class data (C-1.1, C-1.4). Acquirium treats operator logs, maintenance notes, and calibration records as first-class data linked to graph nodes. Each record carries a *record time* and an optional *observation period*, supporting queries such as "all maintenance notes for this pump during last month's outage" and letting applications fold operational context into downstream analytics and quality checks.

Laboratory measurements with richer time semantics (C-1.1, C-1.3). Laboratory assays often have distinct *sample collection* and *analysis* times, unlike inline sensors that produce a single timestamp per reading. Acquirium aligns lab results with operational telemetry by either timestamp as appropriate. This avoids a simplified sensor time-series abstraction that would discard important temporal context.

Portable applications across plants (C-2.1). Acquirium applications are written against semantic graph patterns (e.g., "tank connected to a downstream pump with flow rate property") rather than site-specific tag schemes. An application or ComP implemented once deploys to any plant modeled with WaTr, eliminating the need to rewrite data access logic for each new facility.

Resilience to plant evolution (C-2.2). Changes to plant equipment and instrumentation are reflected in the KG, triggering reevaluation of ComP query objects so that derived property computations dynamically adjust to the current plant state. This reduces the maintenance burden that otherwise grows as plants evolve, since physical changes no longer require manual propagation across application code.

5.6 Implementation

Acquirium is an open source platform implemented in Python. It uses OxiGraph [52] as the RDF graph store and PostgreSQL 17 with the TimescaleDB [69] extension for time series data management. The Acquirium server is implemented as a FastAPI service and users interact via the API or client library. ComP instances run in separate Docker containers, though we are exploring more scalable designs.

6 Evaluation

We evaluate WaTr and Acquirium by measuring their coverage of common system designs (§ 6.1), their coverage of the water treatment domain (§ 6.2) and support of realistic data-driven applications (§ 6.3), and the performance of delivering these features (§ 6.4 and § 6.5).

6.1 Example Models

Table 3 summarizes six example models derived from five WTS. Model 1 is a simple simulation with a single pump. Model 2 represents a wastewater treatment system without any data connections. Model 3 models the Benicia Wastewater Treatment Plant, adapted from [16]. Model 4 extends Model 3 with additional sensors representing a possible retrofit. Model 5 represents a direct potable reuse trailer [17, 40, 43], a water recycling system, created from a P&ID, and verified by the system designer. Model 6 represents the ultrafiltration subsystem of Model 5. These models demonstrate that WaTr concepts are sufficient to model a variety of treatment system architectures and technologies.

Table 3: Summary statistics of RDF models.

Model	#Triple	#Entity	Equip.	ExtRefs	#Conn	#ConnPts
Model-1	63	27	1	5	0	2
Model-2	521	115	18	0	21	37
Model-3	1330	283	21	28	26	48
Model-4	1753	462	21	100	26	48
Model-5	5898	1266	115	112	185	369
Model-6	1651	346	38	24	59	135

Table 4: WaTr coverage of water treatment concepts.

	Fully Represented (%)	Fully + Partially Represented (%)
SDWIS Treatment Objectives	50	100
SDWIS Treatment Processes	34	78
WaterTAP flowsheets	76	100

6.2 Water Domain Concept Coverage

We measure WaTr coverage against two authoritative references: the Safe Drinking Water Information System (SDWIS) Federal Data Reporting Guidelines [71], which enumerate water treatment objectives and processes, and the WaterTAP flowsheets [8, 77], which include unusual configurations (e.g., One-Stack Electrodialysis) that stress WaTr's ability to generalize to unseen designs.

Table 4 reports coverage in two tiers. *Fully* represented concepts have a 1-to-1 correspondence with a WaTr graph; *partially* represented ones are expressible only at a coarser granularity (e.g., SDWIS's "manganese removal" maps to WaTr's generic "heavy metal removal"). At the partial-or-better level, WaTr covers all SDWIS treatment objectives and WaterTAP flowsheets, and 78% of SDWIS processes. Remaining gaps are narrow vocabulary additions planned for the next release.

6.3 Real-World Application Support

Acquirium supports writing and deploying data-driven analytics applications for a wide variety of WTS. We implement five representative WTS applications and compare lines of code (LOC) against best-effort implementations in three comparable data/metadata systems: Mortar [22], Energon [29], and Chrontext [10]. Our evaluation focuses on whether these applications are expressible in WaTr and Acquirium and, if so, the development effort required. All implementations target equivalent functionality over the same KG.

Dataset Construction (DC). Many applications in Table 1 (A1, A4, A5, A7, and A9) require retrieving raw time-series data and assembling structured datasets prior to analysis. We implemented three representative applications to evaluate DC: **(i)** retrieval of tanks and downstream pumps with associated volume and flow

Table 5: i) Comparing development effort across Acquirium, Mortar [22], Energon [29] and Chrontext [10] using lines-of-code metric and ii) Measuring portability of apps

Application	**Acquirium**	[22]	[29]	[10]	Portability
DC-i	**4**	15	5	16	67%
DC-ii	**4**	13	5	11	83%
DC-iii	**4**	17	5	15	17%
CLM	**16**	28	22	29	33%
FDI	**14**	27	14	27	83%

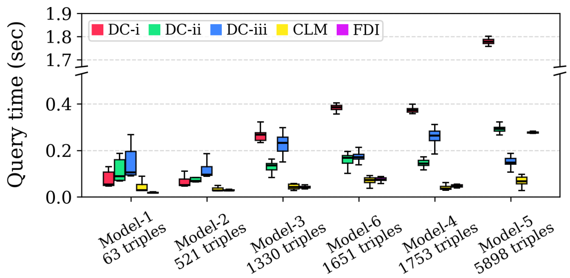
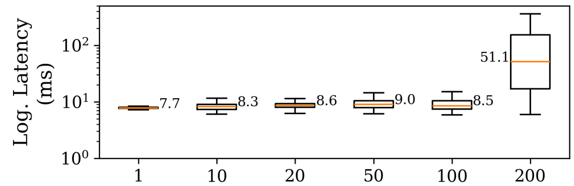
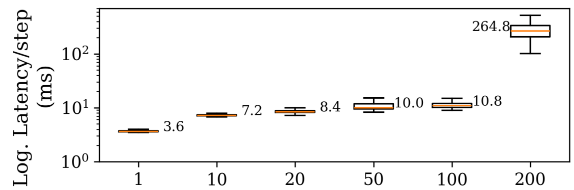


Figure 7: Query latency per model. Note the break in the y-axis.



(a) Per-app latency vs. concurrent ComPs (log y-axis)



(b) Latency vs. ComP chain depth (log y-axis)

Figure 8: Scalability evaluation of ComP execution time.

rate (Fig. 5); **(ii)** influent train quality signals starting from a pump and collecting downstream properties; and **(iii)** secondary clarifier comparison, which retrieves sedimentation tanks with specific roles, their properties, and upstream bioreactors.

Chlorine Level Monitoring (CLM). This application implements real-time CLM in the chlorination basin by periodically retrieving the latest measurements, evaluating them against thresholds, and generating alerts. Fig. 6 illustrates how ComP implementations separate data access, application logic, and output handling.

Fault Detection and Isolation (FDI). We implement an FDI workflow based on [37] that retrieves an in-control reference dataset, applies the FDI algorithm to a test period, and produces a fault report. This evaluates each platform's ability to support complex analytical workflows requiring coordinated access to multiple time windows and datasets.

Results. LOC measurements exclude imports, boilerplate, and visualization code to focus on core application logic and data access. Acquirium consistently requires fewer lines across all applications (Table 5-i). Energon achieves comparable LOC in some cases but relies on a custom declarative query language that limits flexibility, while Mortar and Chrontext incur substantially higher overhead from explicit SPARQL queries and glue code for data retrieval and restructuring. We further measured the fraction of app queries returning results across models (Table 5-ii). Most applications port across plants without code changes; some queries correctly return no values when components are absent (e.g., Model-5, a mobile recycler, lacks the bioreactors of Model-4).

6.4 Graph Query Latency

To quantify graph query latency, we ran the queries from § 6.3 on each model in Table 3, with the WaTr ontology (1680 triples) also inserted into the graph store. Figure 7 reports the results. Graph query latency is largely consistent across models and queries, though some queries (e.g., DC-i) run slower on larger graphs (e.g., Model-5).

6.5 Scalability of Computed Properties

To be effective, Acquirium must support large numbers of applications and soft-sensors while maintaining low processing latency. We evaluate scalability by varying (i) the number of concurrent ComPs and (ii) the depth of composed ComPs, and characterizing the resulting latency distributions. All experiments use Model-4 from Table 3. The workload produces observations at 1 Hz, and each ComP executes a 1 Hz loop that reads its latest input and emits an output. We run the experiments on a cloud VM with 8 vCPUs and 32GB RAM.

We define per-app processing latency as the elapsed time between when an input measurement is received by the ComP and when the ComP completes processing and emits its output message. Both timestamps are recorded within the application process, capturing application-level overhead and excluding upstream sensing delays and network latency.

6.5.1 Scalability of read-only ComP. We deploy concurrent instances (between 1 and 200) of the same lightweight threshold-style warning app. To isolate platform overhead from application logic, each instance performs minimal computation: it reads the latest value for a single point and triggers a message that includes its processing timestamps.

The latency remains stable up to 100 concurrent apps (median $\approx$ 8 ms) and increases moderately at 200 apps (median 51.1 ms). Figure 8a indicates that Acquirium supports highly concurrent workloads with low per-app overhead.

6.5.2 Scalability of read-write ComP. Acquirium supports soft-sensor composition: ComP instances can produce derived time-series that are consumed by other ComPs. To quantify soft-sensor overhead, we construct a linear chain of soft-sensors with depth d. The first ComP reads and increments the raw signal; each subsequent ComP reads and increments the previous ComP's value. Each ComP writes its value back to Acquirium, incurring additional overhead.

We report average ComP execution latency as the number of concurrent soft-sensors increases (Figure 8b). Increasing the number of soft-sensors increases latency as expected: the median rises from 3.6 ms at depth 1, to 10.8 ms at depth 100 and 264.8 ms at depth 200. These results suggest that large numbers of dependent soft-sensors remain practical.

7 Discussion and Limitations

Our evaluation shows that Acquirium supports representative data processing and soft-sensor applications for WTS. WaTr graphs simplify configuration and data discovery for developers while imposing minimal overhead on system scalability. An ongoing extension, "stateful ComPs", persists execution state (e.g., ML model weights) to speed up repeated invocations such as inference.

However, limitations remain and motivate future work. The incremental query API removes the need to learn RDF, SPARQL, or the graph structure: queries use traversal patterns (upstream, downstream) and filter by a small set of attributes (unit, substance, quantity kind), narrowing what a user must reason about at each step. It still assumes familiarity with Python, limiting adoption among non-developer stakeholders. A faceted search interface [7] or LLM code generation against the dataframe abstraction (which LLMs handle better than graph queries [48]) could close this gap.

Although WaTr and Acquirium simplify metadata consumption, building and maintaining these KGs remains a significant adoption barrier. Metadata construction has been a recognized challenge in the buildings domain for over a decade, with early approaches relying on human-in-the-loop techniques [14, 38] and tools such as BuildingMotif [23] reducing manual effort; one of the authors, a water domain expert, took about 8 hours to construct Model 5 (Table 3) using BuildingMotif. LLM-based approaches have recently produced accurate semantic models with minimal supervision [53]. Applying these techniques to WaTr, along with vision-based metadata extraction from P&IDs, is ongoing work.

8 Conclusion

In this work, we co-design a metadata ontology and a data management platform for water treatment operations. The proposed ontology captures a broad spectrum of water treatment technologies and domain concepts, while the platform supports intuitive metadata exploration, data access, and the development of data-driven applications and soft sensors without requiring expertise in semantic query languages. We will continue to improve the performance of Acquirium and the domain coverage of WaTr to support more WTS processes and extend the platform to unstructured data modalities such as text and images that are prevalent in water treatment operations [74].

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